

```
import pandas as pd
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
from google.colab import files
uploaded = files.upload()
```

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```
import pandas as pd
df =pd.read_csv("students.csv")
```

```
df.head()
```

	Weight	Height	Age	Gender
0	79	1.80	35	Male
1	69	1.68	39	Male
2	73	1.82	25	Male
3	95	1.70	60	Male
4	82	1.87	27	Male

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   Weight  10 non-null      int64
 1   Height  10 non-null      float64
 2   Age     10 non-null      int64
 3   Gender  10 non-null      object
dtypes: float64(1), int64(2), object(1)
memory usage: 452.0+ bytes
```

```
df.describe()
```

	Weight	Height	Age
count	10.000000	10.000000	10.000000
mean	72.600000	1.701000	40.300000
std	10.834103	0.118832	22.181323
min	55.000000	1.500000	16.000000
25%	69.000000	1.647500	25.500000
50%	70.000000	1.690000	37.000000
75%	77.500000	1.795000	49.500000
max	95.000000	1.870000	89.000000

```
df["Gender"] = df["Gender"].map({"Male": 1, "Female": 0})
```

```
df
```

	Weight	Height	Age	Gender
0	79	1.80	35	1
1	69	1.68	39	1
2	73	1.82	25	1
3	95	1.70	60	1
4	82	1.87	27	1
5	55	1.55	18	0
6	69	1.50	89	0
7	71	1.78	42	0
8	64	1.67	16	0
9	69	1.64	52	0

```
X=df[["Height", "Age", "Gender"]]
y=df["Weight"]
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
```

```
from sklearn.linear_model import LinearRegression

model = LinearRegression()

model.fit(X_train, y_train)
```

LinearRegression ⓘ ?

```
LinearRegression()
```

```
y_pred = model.predict(X_test)

print("Predicted Weights:")
print(y_pred)
```

Predicted Weights:
[56.85651025 79.10576115]

```
from sklearn.metrics import mean_squared_error, r2_score

print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))
```

Mean Squared Error: 76.57792714844265
R2 Score: -11.252468343750825

```
print("Intercept:", model.intercept_)

print("Height Coefficient:", model.coef_[0])
print("Age Coefficient:", model.coef_[1])
print("Gender Coefficient:", model.coef_[2])
```

Intercept: -3.0552172392595196
Height Coefficient: 32.83533630032696
Age Coefficient: 0.3172947417608091
Gender Coefficient: 14.623118479712073

```
new_person = pd.DataFrame({
    "Height": [1.75],
    "Age": [24],
    "Gender": [1] # Male = 1, Female = 0
})

predicted_weight = model.predict(new_person)

print("Predicted Weight:", predicted_weight[0], "kg")
```

Predicted Weight: 76.64481356828415 kg

```
new_person = pd.DataFrame({
    "Height": [1.8],
    "Age": [35],
    "Gender": [1]    # Male = 1, Female = 0
})

predicted_weight = model.predict(new_person)

print("Predicted Weight:", predicted_weight[0], "kg")
```

Predicted Weight: 81.7768225426694 kg

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